PNAS RESEARCH ARTICLE | APPLIED MATHEMATICS OPEN ACCESS

Sparse autoencoders uncover biologically interpretable features in protein language model representations

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Contributed by Bonnie Berger; received April 20, 2025; accepted July 7, 2025; reviewed by Nir Ben-Tal and Teresa M. Przytycka

Foundation models in biology—particularly protein language models (PLMs)—have enabled ground-breaking predictions in protein structure, function, and beyond. However, the "black-box" nature of these representations limits transparency and explainability, posing challenges for human–AI collaboration and leaving open questions about their human-interpretable features. Here, we leverage sparse autoencoders (SAEs) and a variant, transcoders, from natural language processing to extract, in a completely unsupervised fashion, interpretable sparse features present in both protein-level and amino acid (AA)-level representations from ESM2, a popular PLM. Unlike other approaches such as training probes for features, the extraction of features by the SAE is performed without any supervision. We find that many sparse features extracted from SAEs trained on protein-level representations are tightly associated with Gene Ontology (GO) terms across all levels of the GO hierarchy. We also use Anthropic's Claude to automate the interpretation of sparse features for both protein-level and AA-level representations and find that many of these features correspond to specific protein families and functions such as the NAD Kinase, IUNH, and the PTH family, as well as proteins involved in methyltransferase activity and in olfactory and gustatory sensory perception. We show that sparse features are more interpretable than ESM2 neurons across all our trained SAEs and transcoders. These findings demonstrate that SAEs offer a promising unsupervised approach for disentangling biologically relevant information present in PLM representations, thus aiding interpretability. This work opens the door to safety, trust, and explainability of PLMs and their applications, and paves the way to extracting meaningful biological insights across increasingly powerful models in the life sciences.

interpretability | protein language models | sparse autoencoders | transcoders

Significance

Interpreting representations derived from protein language models (PLMs) is crucial for improving trust in the model, explainability, and human–AI collaboration in downstream applications. We leverage sparse autoencoders and transcoders from natural language processing as a way to extract biologically meaningful, interpretable features from both protein-level and amino acid-level representations. Our Gene Ontology Analysis and automated interpretability protocols uncover many sparse features that are strongly associated with specific functional annotations and protein families. We show that the sparse features are more interpretable than PLM neurons. These insights not only enhance our understanding of biological information PLMs encode and provide a pathway for gaining functional meaning from PLMs, but also enable interpretability for downstream tasks that rely on these representations.

Protein language models (PLMs) (1–3) are trained in an unsupervised manner on large datasets of protein sequences. A popular approach is through masked language modeling, where specific tokens are masked and the model aims to predict the masked token (1, 2). After pretraining in this unsupervised fashion, protein-level representations from these models are frequently extracted for use on downstream protein-related tasks (4, 5). Transformer-based PLMs like ESM2 (2) commonly yield $n \times d$ shaped representations for any layer, where n is the length of the protein sequence and d is the size of the model's vector for each position of the sequence. The protein-level representations are typically derived through mean-pooling (over the sequence dimension n) of the amino acid-level (token-level) representations (4), effectively condensing sequence-level information into a (fixed-length) d-dimensional vector for each protein. Although this approach facilitates efficient downstream applications, these representations are hard to interpret.

This lack of interpretability creates a "black-box" problem: Protein-level representations may yield high accuracy on downstream tasks, but provide little insight into what biological features are encoded in each representation and why it leads to specific downstream predictions. Understanding these representations is crucial to explainability and fostering human–AI collaboration in such downstream tasks. On the other hand, the ability to interpret amino acid-level (token-level) representations would provide a window into the internal operations of PLMs and uncover biological attributes that these models track. As PLMs continue to grow more powerful, the ability to make sense of both protein-level and amino acid-level representations will become increasingly valuable for both understanding downstream predictions and extracting meaningful biological insights, as well as ensuring that open-sourcing these models does not inadvertently expose potentially harmful biological information.

Author contributions: O.G., M.B., E.A., and B.B. designed research; performed research; contributed new reagents/analytic tools; analyzed data; and wrote the paper.

Reviewers: N.B.-T., Tel Aviv University; and T.M.P., NIH.

The authors declare no competing interest.

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This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2506316122/-/DCSupplemental.

Published August 19, 2025.

Several studies have sought to understand how PLMs encode biological information. One line of exploration has shown that models like ESM2 predict protein structures by leveraging coevolutionary statistics of interacting motifs, instead of explicitly having captured the physics of protein folding (6). Another approach has analyzed transformer-based protein models by assessing how attention heads and internal embeddings capture specific features, such as contact maps and binding sites, showing that deeper layers of the model capture more complex properties (7). Although these studies provide insights into how PLMs generally store information and the kind of information they contain, they do not offer an unsupervised method to extract the biologically meaningful features hidden in a particular PLM representation. We aimed to do exactly that, by leveraging an unsupervised approach that systematically disentangles individual PLM representations (at both the amino acid and protein-level) into interpretable features. An unsupervised approach is desirable with this kind of interpretability, as one may not always know in advance what to look for.

We draw inspiration from a very recent breakthrough in the natural language setting with the introduction of sparse autoencoders. A challenge in neural network interpretability is that neurons in general (8) [and in language models in particular (9–11)] are polysemantic: Each neuron is activated by multiple, unrelated features. The reason behind this is that the features they represent in the real world occur very sparsely, so there are a lot more distinct features that the model needs to express than there are neurons in a layer. So instead of dedicating a full neuron to each sparsely occurring feature, a layer of a neural network tries to accommodate a lot more sparsely occurring features than it has neurons in the layer by storing these features in superposition. Although this leads to a well-performing neural network, it has proven to be a significant challenge for interpretability. Passing a natural language model's internal representation through a sparse autoencoder was shown to help extract interpretable sparse features from the model's representation in an unsupervised manner (9–13).

In this work, we apply sparse autoencoders (SAEs) to PLMs to interpret PLM representations. We train SAEs on both protein-level and amino acid-level representations extracted from multiple layers of ESM2 and analyze the sparse features (i.e. neurons of the SAE's hidden layer) they uncover. To interpret these features, we conduct Gene Ontology (GO) enrichment analysis on proteins that strongly activate specific sparse features in SAEs trained on protein-level representations and identify meaningful associations with diverse biological functions, from fundamental metabolic pathways like iron-sulfur cluster assembly to specialized functions such as viral translational frameshifting.

Once the sparse features have been extracted in an unsupervised manner by our SAE, we also leverage a scalable, automated large language model (LLM)-assisted interpretability approach, as opposed to humans, to interpret the sparse features. We find that many sparse features correspond to biologically meaningful entities like protein families and functions, which include the NAD Kinase family involved in NAD metabolic processes and NADP biosynthesis, the IUNH family linked to nucleoside hydrolase activity, and the PTH family. Notably, sparse features are more interpretable than standard ESM2 neurons across all tested layers.

Further, we introduce transcoders (14) to PLMs to learn a sparsified approximation of the transformation from the protein-level representation in one layer of ESM2 to the next, with interpretable sparse features. These transcoder-derived sparse features exhibit interpretability on par with those from SAEs, suggesting that they offer an alternative way to understand the organization of information within individual PLM representations. Collectively, our findings demonstrate that SAEs and their variants disentangle complex PLM representations into biologically interpretable features in an unsupervised fashion.

By disentangling biologically relevant features in an unsupervised manner from protein-level representations, our approach has broad implications for explainability in the context of downstream protein-level tasks. At the same time, this approach also opens up the potential to gain fresh biological insights from systematically understanding interpretable features that PLMs track in AA-level representations. Moreover, as foundational models get more advanced, we expect SAEs and their variants to play an increasingly significant role in their interpretability.

Results

Overview of Method.

Sparse autoencoders. A sparse autoencoder (9–13, 15, 16) is an autoencoder, with a single hidden layer, but instead of the hidden layer being much narrower than the input, it is much wider; this autoencoder is constrained to activate the neurons of the hidden layer in a sparse fashion on any given input. A sparse autoencoder is trained on the activations of a chosen layer of a model. The result is that the wide hidden layer has sparsely activating neurons, thereby disentangling polysemantic neurons into sparse neurons that, in the NLP setting, have been shown to be more monosemantic (therefore more interpretable) than the original neurons. These sparsely activating neurons are referred to as sparse features or SAE features throughout the paper. Several groups have used sparse autoencoders to interpret large language models trained on natural language (9–13). They found several sparse features that capture interpretable features in natural language. For instance, they find a feature that activates on Hebrew script, another feature that captures DNA sequences, and yet another one that captures mathematical symbols (9).

Our workflow. Inspired by the success of sparse autoencoders in the NLP setting, here we leverage sparse autoencoders to interpret PLMs. We train SAEs to interpret i) protein-level representations and ii) amino acid-level representations from various layers of the ESM2 model: "esm2 t12 35M UR50D." Fig. 1*A* illustrates the two points in the process where we may attempt to interpret PLM representations. The first is interpreting the protein-level representations from ESM2, which is helpful from a downstream perspective since these representations get used on downstream protein-level tasks. The second is interpreting the amino acid-level representations from ESM2, which is helpful for understanding what features ESM2 seems to be tracking.

Although there are multiple ways to extract protein-level representations from ESM2, a very commonly used and often superior way tends to be mean-pooling (4), so we employ it in this work. We trained SAEs (schematic in Fig. 1*B*) to extract features (in an unsupervised fashion) from the protein-level representations and amino acid-level representations from layers 6, 7, 8, 9, 10 of ESM2, and then aimed to interpret these sparse features in the context of the following metadata: protein names, protein families, gene names, and GO terms, including biological processes, cellular components, and molecular functions. See *Materials and Methods* for more details on the sparse autoencoder architecture and interpretation methodology.

Transcoders. We also train a variant of sparse autoencoders called transcoders [originally proposed by Dunefsky et al. in the NLP

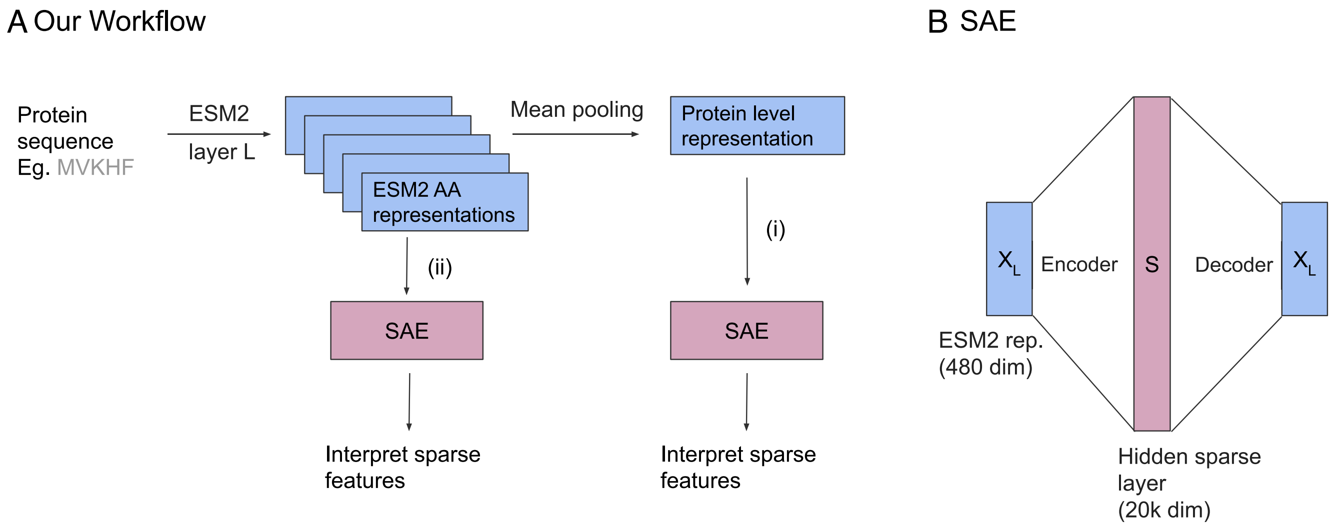

Fig. 1. Interpreting ESM representations with SAEs. (*A*) Brief illustration of our workflow highlighting the two places at which we choose to interpret representations from ESM2 with the help of a sparse autoencoder (SAE): i) at the protein level by passing the single protein-level representation through the SAE—since these representations get used on a lot of downstream tasks, interpreting them is helpful for human–AI collaboration and explaining performance on downstream tasks. ii) at the amino acid-level, by passing individual amino acid (AA) level representations through the SAE one at a time, allowing us to understand what features ESM2 is tracking. The blue boxes are representations drawn from the ESM2 model (at the protein-level or AA-level respectively); these are represented as X_L in (*B*). (*B*) Schematic of the sparse autoencoder (SAE) with a hidden dimension of 20,000 trained on X_L, which is either the 480-dimensional protein-level representation, or 480-dimensional individual AA-level representation from layer L of ESM2.

setting (14)] on the protein-level representations. Here, the input and sparsity constraint on the hidden layer is the same as for our sparse autoencoder, but the output that it aims to predict is the protein-level representation of the next layer of the PLM. So these transcoders are made to learn a sparse approximation of the computation to go from one layer's protein-level representation to the next. The resulting sparse features are interpreted similarly to those from our SAEs.

We trained sparse autoencoders and transcoders on protein-level and amino acid-level representations extracted from layers 6, 7, 8, 9, 10 of ESM2.*

GO Analysis Uncovers Strong Associations of Sparse Features with GO Terms. To investigate the biological interpretability of the sparse features learned from the SAEs trained on protein-level representations, we aimed to analyze their association with functional annotations. We work with a dataset of 18,142 randomly selected human proteins from the UniProt dataset (17) for this analysis.† For each sparse feature, we identified the set of proteins in our dataset for which the feature was activated. All proteins with a nonzero activation value for a specific feature were included in the candidate set of proteins for that feature.

We performed a GO (18) enrichment analysis for each set of highly activated proteins. GO annotations provide a standardized vocabulary to describe the molecular functions, biological processes, and cellular components associated with proteins. Using the set of proteins corresponding to a sparse feature, we tested for overrepresentation of GO terms relative to the background set of all proteins in the dataset. Enrichment analysis was conducted using a hypergeometric test, and *P*-values were corrected for multiple hypothesis testing using the Benjamini–Hochberg procedure to control the false discovery rate.

The results of the GO enrichment analysis revealed that certain sparse features were strongly associated with niche functional annotations. For instance, one feature was highly activated in proteins enriched for "retrotransposition" (GO:0032197) and "transposition" (GO:0032196), while another feature was associated with "iron-sulfur cluster assembly" (GO:0016226) and "metallo-sulfur cluster assembly" (GO:0031163), functions characteristic of specific metabolic processes. Importantly, many of these enriched terms represented distinct biological functions or processes, such as "pentose-phosphate shunt, oxidative branch" (GO:0009051) and "positive regulation of protein oxidation" (GO:1904808), suggesting that the sparse autoencoder successfully disentangled biologically meaningful patterns from the original protein-level ESM representations.

To further assess the interpretability of the sparse features, we aimed to evaluate two key properties of the protein sets associated with each feature. First, we sought to determine whether certain features activated only for very specific functions, as indicated by GO terms that were proximal in the GO Directed Acyclic Graph (DAG) (18). Such specificity would not only imply that the feature was capturing meaningful functional attributions for the proteins that activated it, but also suggest that the sparse feature was learning a more "monosemantic" representation; by this we mean a feature that favors GO terms that are closer in the GO DAG and use that as a proxy for GO monosemanticity here. To quantify this, for each sparse feature, we took the set of GO terms it enriched (with significance $\leq$0.05) and computed the pairwise shortest path lengths between all terms in the original GO DAG. The mean shortest path length was then calculated per sparse feature. A smaller mean shortest path length indicated that the enriched GO terms were more proximal in the DAG, reflecting a higher degree of GO monosemanticity for the feature.

After understanding the GO monosemanticity of each sparse feature, we then wanted to assess where on the GO DAG these monosemantic features mapped. To do this, for each sparse

*The motivation for interpreting these layers of ESM2 is that the ESM2 model we work with has a total of 12 layers, and drawing on intuition from interpretability work on LLMs (10), we would expect the most interesting layers for interpretation to be those somewhere near the end of the model, however not absolutely at the end—since those are likely to be have been specialized for the unsupervised model training task.

†In total there are 26467, 10146, 4022 GO terms in the GO directed acyclic graph (DAG) belonging to the 3 namespaces: biological process, molecular function, and cellular component, resp. Of these, 11545, 4354, 1713 GO terms appear in our set of GO terms used in the protein set used for our GO analysis. Thus, for the 3 namespaces, 43.62%, 42.91%, 42.59% of the GO terms, resp., are covered by our protein set used for the GO analysis.